

Practical No. 10

Website Hosting

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Batch - S5

Study and Implementation of Web Application Deployment

Problem Statement 1: Basics of Deployment

1. What is web application deployment, and why is it necessary?

Ans -

Web application deployment is the process of making a web application available on the internet so users can access it using a URL.

It is necessary because:

- Local apps are only accessible on your machine
- Deployment makes the app accessible globally
- Enables real-world usage and testing

2. Differentiate between local development and production deployment.

Ans -

Feature	Local Development	Production
Environment	Local machine	Cloud server
Access	Only developer	Public users
Database	Local DB	Cloud DB
Performance	Not optimized	Optimized
Security	Minimal	High

3. Explain the role of a web server (e.g., Apache, Nginx) in deployment.

Ans -

A web server:

- Handles incoming HTTP requests
- Serves static files (HTML, CSS, JS)
- Acts as reverse proxy
- Improves performance and security

Example:

- Nginx routes traffic to backend (Node.js)

4. What are environment variables, and how are they used in deployment?

Ans -

Environment variables store configuration values like:

PORT=5000

MONGO_URI=...

JWT_SECRET=...

Used for:

- Security (hide secrets)
- Different configs for dev & production

5. How do you configure a Node.js/Express.js application for deployment?

Ans -

Steps:

- Use `process.env.PORT`
- Use `.env` for local, dashboard ENV for cloud
- Add start script:

```
"scripts": {  
  "start": "node index.js"  
}
```

6. What is the difference between manual deployment and automated deployment (CI/CD)?

Ans -

Type	Description
Manual	Developer deploys manually
CI/CD	Automated deployment on push

Example:

- GitHub → Auto deploy on Render

7. Explain the concept of hosting services (Heroku, Netlify, Vercel, AWS, etc.).

Ans-

Platform	Use
Render	Full-stack apps
Vercel	Frontend
Netlify	Static sites
AWS	Enterprise
Heroku	Easy deployment

8. What are the common challenges faced during deployment, and how can they be resolved?

Ans -

Issue	Solution
DB not connecting	Use cloud DB (MongoDB Atlas)
Env missing	Add in hosting dashboard
CORS error	Configure backend
Port issue	Use <code>process.env.PORT</code>

9. How do you ensure security (HTTPS, authentication, environment secrets) in deployment?

Ans -

Use HTTPS

- Store secrets in ENV
- Use authentication (JWT)
- Avoid exposing `.env`

10. What is the importance of logging and monitoring in deployed applications?

Ans -

Helps debug errors

- Track user activity
- Tools:
 - Render logs
 - Console logs
 - Monitoring tools

Problem Statement 2: Deploying a Basic Express.js Application

Requirements

- Create a simple Express.js application with routes:
 - o GET / → Responds with "Application Deployed Successfully".
 - o GET /**status** → Responds with a JSON object **{status: "running", environment: "production"}**.

Code -

```
import express from "express";
import dotenv from "dotenv";

dotenv.config();

const app = express();

app.get("/", (req, res) => {
  res.send("Application Deployed Successfully");
});

app.get("/status", (req, res) => {
  res.json({
    status: "running",
    environment: process.env.NODE_ENV || "production"
  });
});

const PORT = process.env.PORT || 5000;

app.listen(PORT, "0.0.0.0", () => {
  console.log(`Server running on port ${PORT}`);
});
```

- Configure the application to run on a specified port using environment variables.

My Workspace

My project / Production / New Web Service

Search

+ New Upgrade

New Web Service

Source Code

Atharv-K-979 / WCE-Labs-Material · 1h ago

Edit

Name

WCE-Labs-Material-1

Project Optional

Add this web service to a project once it's created.

My project

/

Production

Language

Choose the runtime environment for this service.

Node

Branch

The Git branch to build and deploy.

main

Region

Your services in the same region can communicate over a private network. You currently have services running in **Oregon**.

Oregon (US West)

1 existing service

Deploy in a new region +

Root Directory Optional

If set, Render runs commands from this directory instead of the repository root. Additionally, code changes outside of this directory do not trigger an auto-deploy. Most commonly used with a monorepo.

./src

Build Command

\$ yarn

- Deploy the application on a free hosting platform (e.g., Heroku, Render, or Vercel).

Root Directory Optional

If set, Render runs commands from this directory instead of the repository root. Additionally, code changes outside of this directory do not trigger an auto-deploy. Most commonly used with a monorepo.

/Web-Dev-Design-Lab/Assignment10

Build Command

Render runs this command to build your app before each deploy.

/Web-Dev-Design-Lab/Assignment10/ \$ npm install

Start Command

Render runs this command to start your app with each deploy.

/Web-Dev-Design-Lab/Assignment10/ \$ npm start

Instance Type

For hobby projects

Free

\$0 / month

512 MB (RAM)

0.1 CPU

Upgrade to enable more features

Free instances spin down after periods of inactivity. They do not support SSH access, scaling, one-off jobs, or persistent disks. Select any paid instance type to enable these features.

For professional use

For more power and to get the most out of Render, we recommend using one of our paid instance types. All paid instances support:

- Zero Downtime
- SSH Access
- Scaling
- One-off jobs
- Support for persistent disks

Starter

\$7 / month

512 MB (RAM)

0.5 CPU

Standard

\$25 / month

2 GB (RAM)

1 CPU

Pro

\$85 / month

4 GB (RAM)

2 CPU

Pro Plus

\$175 / month

8 GB (RAM)

4 CPU

Pro Max

\$225 / month

16 GB (RAM)

4 CPU

Pro Ultra

\$450 / month

32 GB (RAM)

8 CPU

Need a custom instance type? We support up to 512 GB RAM and 64 CPUs.

Environment Variables

Set environment-specific config and secrets (such as API keys), then read those values from your code. [Learn more.](#)

PORT

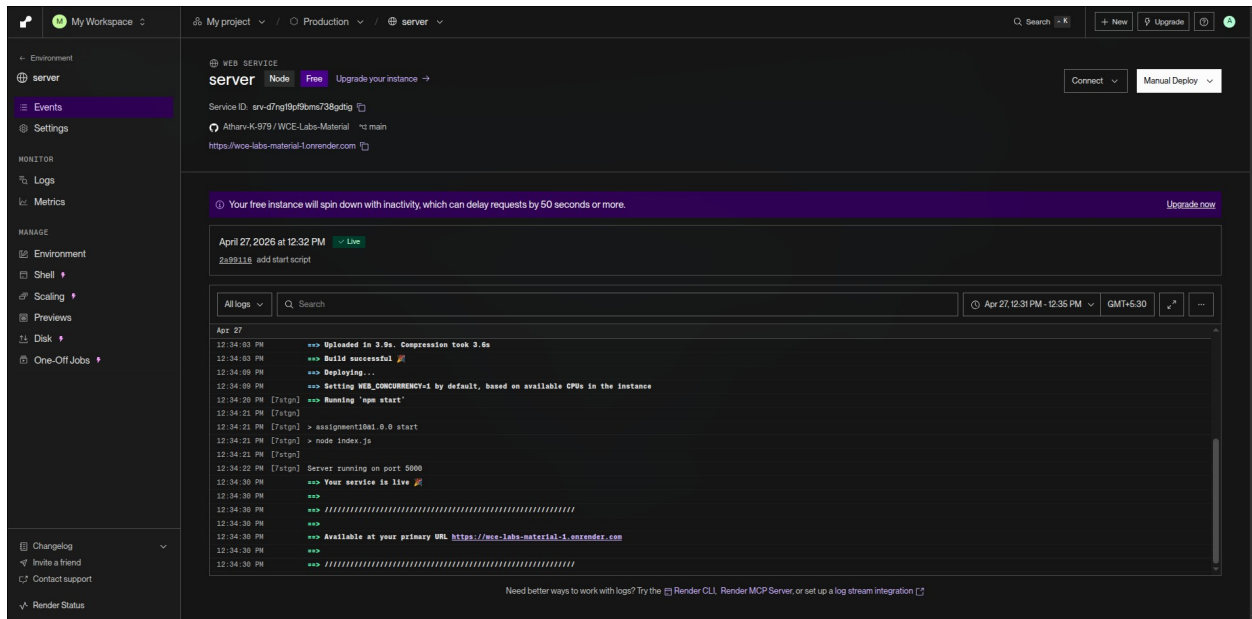
.....

NODE_ENV

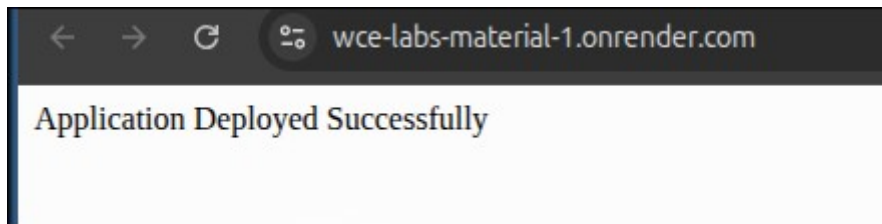
.....

+ Add Environment Variable

Add from .env



- Verify deployment by accessing the live URL in a browser.



Problem Statement 3: Advanced Deployment Features

- Add a route:
 - GET **/info** → Responds with application details (name, version, author).
- Configure error handling for invalid routes (404).
- Implement logging middleware to track requests.

```
import express from "express";
import dotenv from "dotenv";

dotenv.config();

const app = express();

//Logging Middleware
app.use((req, res, next) => {
  const time = new Date().toISOString();
  console.log(`[${time}] ${req.method} ${req.url}`);
  next();
});

// Home
app.get("/", (req, res) => {
  res.send("Application Deployed Successfully");
});

// Status
app.get("/status", (req, res) => {
  res.json({
    status: "running",
    environment: process.env.NODE_ENV || "development"
  });
});

// Required: /info route
app.get("/info", (req, res) => {
  res.json({
    name: "Blog Application",
    version: "1.0.0",
    author: "Atharv Kulkarni"
  });
});

app.use((req, res) => {
  res.status(404).json({
    error: "Route not found"
  });
});
```

- Demonstrate environment-specific configuration (development vs. production).

Environment	Behavior
development	Detailed logs, local testing
production	Optimized performance, deployed

- Document the deployment steps (installation, configuration, deployment, verification).

```

Mon 27 Apr - 18:38 ~/Documents/WCE_Lab_Material/Web-Dev-Design-Lab/Assignment10
origin Ω main 3* 3● 81-
@atharv > npm init -y
npm install express dotenv
Wrote to /home/atharv/Documents/WCE_Lab_Material/Web-Dev-Design-Lab/Assignment10/package.json:

{
  "name": "assignment10",
  "version": "1.0.0",
  "main": "index.js",
  "type": "module",
  "scripts": {
    "start": "node index.js"
  },
  "dependencies": {
    "express": "^4.18.2",
    "dotenv": "^16.0.0"
  },
  "devDependencies": {},
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}

added 8 packages, removed 4 packages, changed 22 packages, and audited 71 packages in 4s

17 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

```

```

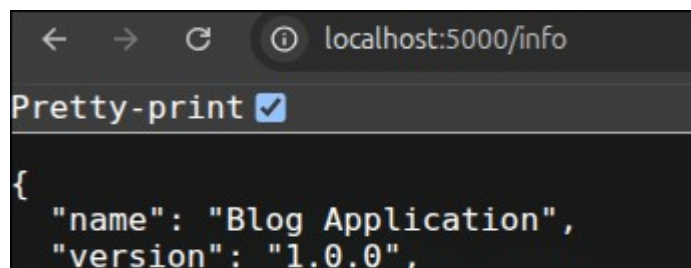
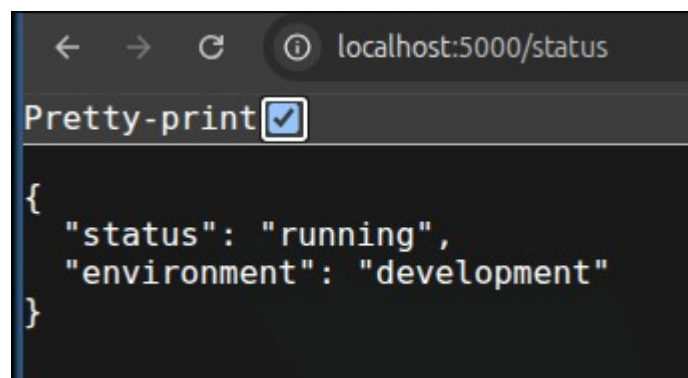
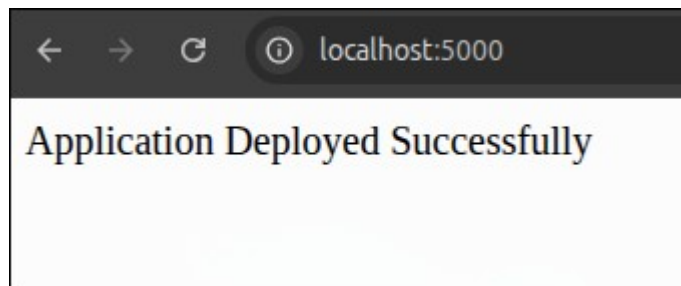
Mon 27 Apr - 18:38 ~/Documents/WCE_Lab_Material/Web-Dev-Design-Lab/Assignment10
origin Ω main 38* 116● 117-
@atharv > npm start

> assignment10@1.0.0 start
> node index.js

Server running in undefined mode on port 5000

```


Deployment Local



Logs

```
Server running in undefined mode on port 5000
[2026-04-27T13:09:35.177Z] GET /
[2026-04-27T13:09:35.346Z] GET /favicon.ico
[2026-04-27T13:10:39.628Z] GET /status
[2026-04-27T13:12:09.208Z] GET /info
```

Deployment Production

Environment Variables

Set environment-specific config and secrets (such as API keys), then read those values from your code. [Learn more.](#)

Export

Edit

KEY	VALUE
NODE_ENV	production
PORT	5000

WEB SERVICE

server

Node

Free

Upgrade your instance →

Connect

Manual Deploy

Service ID: `srv-d7ng19pf9t8ms738gdtig`

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<https://wce-labs-material-1.onrender.com>

ⓘ Your free instance will spin down with inactivity, which can delay requests by 50 seconds or more.

Upgrade now

April 27, 2026 at 6:48 PM Live

`fb4c568` updated `index.js`

All logs

Q Search

Apr 27, 6:47 PM - 6:50 PM

GMT+5:30

⌵

...

Apr 27

06:49:00 PM ==> Deploying...

06:49:00 PM ==> Setting `WEB_CONCURRENCY=1` by default, based on available CPUs in the instance

06:49:11 PM [v7krc] ==> Running `'npm start'`

06:49:12 PM [v7krc] > `assignment1981.0.0 start`

06:49:12 PM [v7krc] > `node index.js`

06:49:12 PM [v7krc]

06:49:14 PM [v7krc] Server running in production mode on port 5000

06:49:14 PM [v7krc] [2026-04-27T13:19:14.241Z] HEAD /

06:49:23 PM ==> Your service is live 🎉

06:49:23 PM ==>

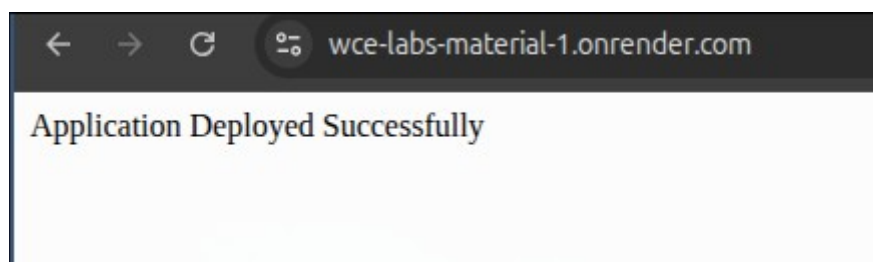
06:49:23 PM ==> Available at your primary URL <https://wce-labs-material-1.onrender.com>

06:49:23 PM ==>

06:49:23 PM ==> /

06:49:23 PM [v7krc] [2026-04-27T13:19:23.465Z] GET /

Need better ways to work with logs? Try the [Render CLI](#), [Render MCP Server](#), or set up a [log stream integration](#)



404 Error Handling (As user is not authenticated he get 404)

